

Heba Alazzez

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ColorStack | Girls Who Code | Rewriting The Code | Salesforce Tech Accelerator | Grow With Google Scholar

EDUCATION

University of California, Berkeley
Computer Science B.A.

Expected Graduation - **May 2027**

College of San Mateo
Computer Science, Physics, and Mathematics A.S.

August 2023 - May 2025

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Object Oriented Programming, Systems Programming, Software Engineering Methods, Discrete Mathematics, Linear Algebra.

EXPERIENCE

Google

May 2026

Software Engineer Intern

- Incoming Google Cloud SDK Platform.

Snap Inc. (Snapchat)

June 2025 – August 2025

Snap Lens Academy Intern

- Developed and deployed **30+** Augmented Reality experiences using **JavaScript** and Lens Studio's API framework, 3D Assets leveraging modular script architectures to ensure performance at scale (**4M+ global users**).
- Selected as **1 of 15 from 300+ applicants** for Snap's AR/VR engineering academy.
- Delivered **10+ live demos** to Snap's CEO, CTO, and engineering teams. [Featured in CEO Evan Spiegel's post](#)

Uber

December 2024 – August 2025

Software Engineer Fellow

- Selected as **1 of 50** from **2,000+ applicants globally (top 3%)** for Uber's international Career Prep Program.
- Solved **50+ advanced algorithmic challenges** curated by Uber Engineering, covering graph traversal (DFS, BFS, Dijkstra), backtracking, greedy, and dynamic programming.
- Strengthened **system design** fundamentals by combining Big-O analysis with API modeling, modular service architecture, and distributed scalability strategies.

Kodely

June 2024 – January 2025

Software Engineer Intern

- Developed responsive enrichment program platform for K–12 districts using **Astro, React, Tailwind CSS, and Firebase**, integrating real-time data sync and authentication to drive a **75%** increase in student engagement.
- Optimized web performance by implementing **code-splitting, lazy loading, and server-side rendering**, reducing average page load times by **40%** across client-facing dashboards.

RESEARCH & PUBLICATIONS

Stanford Intelligent Systems Laboratory

June 2024 – Present

AI Researcher

- Developed a **real-time LLM-powered pilot assistant**, combining **Retrieval Augmented Generation** with aircraft state awareness, trajectory prediction, and natural language interfaces.
- Implemented core AI services: **runway lookup algorithm** (5 nearest airports in <200ms), **METAR/TAF weather parser** (99% accuracy), and **voice pipeline** (95% transcription accuracy).
- Presented results to Airbus, NASA, and U.S. Air Force Test Pilot School; system is **undergoing in-flight testing**. Co-authored publication accepted to the **European Conference on AI 2025**. <https://arxiv.org/abs/2503.16477>

Stanford Mineral-X

June 2024 – July 2025

AI Researcher

- Developed a **Python/Mayavi-based 3D subsurface visualization tool**, enabling **dynamic slicing** and interactive volumetric rendering, improving geological analysis speed by **3x**.
- Implemented modularized architecture with **NumPy** and **PyVista**, optimizing pipelines to process **10M+ voxel geological datasets** with <1s latency on modern GPUs.
- Delivered **codebase and documentation** to the **Mineral-X consortium**, supporting studies across **5+ research teams**.

[Github](#)

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, JavaScript, TypeScript, PHP, HTML/CSS, Julia, LaTeX, MatLab, LangChain

Frameworks: React, Node.js, Next.js, WordPress, GraphQL, Django, Flask, Express.js

Developer Tools: Git, GitHub, GitLab, Docker, Kubernetes, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, JIRA

Libraries: Pandas, NumPy, Matplotlib, Mayavi, SciPy, Seaborn, OpenAI API

Databases: PostgreSQL, MySQL, Firebase, SQLite